Consulting & Analytics Club
IIT Guwahati

Summer **Projects** '26

Problem Statements

- Digital Lending: Portfolio Optimization
- Decoding Customer Value: A SQL-Driven Retention Strategy
- Unlocking Behavioral Intelligence in Airline Loyalty Programs
- Optimizing Delivery ETAs with Graph-Based Network Intelligence

Digital Lending: Portfolio Optimization

Industry Context

Digital lending institutions in emerging markets have witnessed rapid growth by leveraging technology-driven customer acquisition, alternative data sources, and faster credit decisioning. These lenders cater to a diverse customer base across urban and semi-urban regions, including first-time borrowers and underbanked segments, through products such as unsecured personal loans, SME working capital loans, and buy-now-pay-later (BNPL) offerings.

While scale and speed have enabled strong topline growth, evolving borrower behavior, macroeconomic volatility, and rising acquisition costs have increased pressure on portfolio quality and profitability. Many lenders now face the challenge of balancing growth ambitions with prudent risk management, while effectively leveraging the rich data generated across the customer lifecycle.

Business Challenge

Despite having access to extensive customer, loan, repayment, and behavioral data, lenders often struggle to translate this information into actionable insights for strategic decision-making. Key issues include limited visibility into risk drivers, delayed identification of potential delinquencies, suboptimal pricing and product structures, and fragmented performance monitoring for leadership.

The central challenge is to design a data-driven approach that enables lenders to understand customer risk and value holistically, detect early signs of financial stress, optimize growth strategies without compromising portfolio quality, and support leadership with clear, forward-looking insights.

Problem Statement

How can a digital lending institution leverage end-to-end customer and transaction data to strengthen credit risk assessment, enable early detection of delinquencies, optimize pricing and product strategies, and drive sustainable, risk-adjusted portfolio growth?

Key Questions to Address

Participants are expected to explore and answer the following:

1. Which customer segments exhibit materially different risk and repayment behaviors, and what attributes define these segments?
2. How do acquisition channels and onboarding strategies impact portfolio quality, customer lifetime value, and unit economics?
3. Which loan products, ticket sizes, and tenures deliver the strongest balance between growth and risk?
4. How can pricing, approval, or tenure strategies be tailored across segments to improve overall portfolio outcomes?
5. What performance metrics and views should senior leadership monitor to proactively manage risk and growth?

Data Design Guidelines

Participants are required to design and generate their own synthetic dataset that reflects a realistic digital lending ecosystem. The dataset should capture the full customer journey and allow meaningful analysis of risk, growth, and performance across the following areas:

Data Area	Core Information to Include	Expected Relationship
Customer Profile	Geography, income proxy, employment type, credit quality indicator	Weaker profiles tend to exhibit higher risk
Loan & Product	Product type, loan size, tenure, pricing, origination risk grade	Higher risk may result in tighter terms or higher pricing
Repayment Behavior	Payment timeliness, delinquency status, partial payments	Deteriorating payment behavior precedes default
Behavioral Signals	Cash flow consistency, balance volatility, spending shocks	Higher volatility may result in higher delinquency likelihood
Acquisition	Channel, cost of acquisition, approval turnaround	Faster/cheaper growth may carry higher risk
Time Dimension	Origination and repayment periods	Newer cohorts may behave differently
Outcomes	Loan status, default indicator, value proxy	Certain segments outperform on risk-adjusted basis

Deliverables

The final submission consists of two components: a **Credit Policy Recommendation Report** and a **Presentation deck**.

1. The report should be 4 to 6 pages in length, addressed to a fictional Chief Risk Officer. It must include an executive summary, risk segmentation findings, early warning signal logic, and a specific policy recommendation accompanied by a projected impact assessment. This format mirrors what a strategy analyst at a fintech or a consulting associate on a financial services engagement would be expected to produce.
2. The presentation deck should complement the report and communicate the team's solution to a stakeholder audience. Its structure and design are intentionally left to the participants' discretion; we are more interested in the quality of thinking than in adherence to a prescribed format.

Across both components, the standard being applied is this: the work should lead with strategic recommendations that are substantiated by data, not data outputs in search of a conclusion.

Point of Contact

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Decoding Customer Value: A SQL-Driven Retention Strategy

Background

A direct-to-consumer (D2C) fashion brand sells clothing, accessories, footwear, and outerwear across the United States. The brand has no physical stores and no third-party retailers — every customer relationship is managed directly by the brand itself.

The brand has grown steadily and now has customer behavioral data covering around 3,900 customers. It runs a promotional discount program, supports multiple payment methods, and offers a range of shipping options. But it has never built a structured way to understand its customers beyond surface-level sales numbers.

The founding team is at a critical point: they can keep running promotions reactively and hope customers keep coming back — or they can build something more deliberate. They have chosen to build.

Core Problem

The brand has data but no intelligence built on top of it. Specifically, it cannot answer:

- Who are the customers likely to still be buying two years from now, and what do they look like today?
- Is the discount and promo program actually building a loyal customer base, or just attracting one-time bargain hunters?
- Which product categories are associated with lower purchase history, and which ones appear most among high-frequency, high-tenure customers?
- Are there cities or regions where the brand has strong traction that it has not yet deliberately targeted, and does that traction look different in terms of category preference, promo sensitivity, or average spend per customer?
- What does the brand's best customer actually look like in terms of age, purchase habits, payment preferences, and satisfaction?

Without answers to these questions, the brand is making marketing and product decisions based on gut feel rather than evidence.

Problem Statement

Using only transactional and behavioral data, can the brand identify what its most valuable customers look like, measure how much of its current revenue depends on promotions, and build a data-backed retention strategy that reduces discount dependency without hurting sales?

Key Questions to Address

1. Who are the genuinely loyal customers vs. those who only buy when there is a discount?
2. What behavioral patterns today predict high customer value over time?
3. Which geographies and demographics are commercially underlevered?
4. How should the brand restructure its promotional strategy to protect margins without losing volume?
5. What does the brand's ideal customer profile look like, and how can it acquire more of them?

Scope of Analysis

1) Data Preparation & Feature Engineering (Python):

Clean and prepare the raw dataset. Then build the customer-level metrics the brand needs — but here is the constraint: you are not told which metrics to build. You need to decide what to measure and justify why those choices make sense for this business.
As a starting point, think about what signals in the data could indicate value, satisfaction, and promotional reliance. But do not stop at computing numbers, explain the logic behind each metric you create and what it is actually trying to capture.
One thing to keep in mind: metrics that sound analytical but do not lead to a decision are not useful. Every engineered feature should answer a question the brand actually cares about.

2) Customer Segmentation & Analysis (SQL)

Build a structured query layer to answer the brand's core business questions:

- What separates high-value customers from low-value ones, and which profiles show the strongest repeat purchase behavior?
- Which seasons and categories are associated with lower-tenure customers versus those with high previous purchase counts?
- Which geographies signal organic demand versus discount-driven volume?

3) Founder Dashboard (Power BI)

Build a clean, four-panel dashboard designed for a non-technical founding team:
• Customer pyramid: showing how value is distributed across the customer base
• Promo dependency vs. retention rate: plotted by segment to show who needs discounts to buy and who doesn't
• Geographic opportunity map: not just which regions buy most, but which regions show high spend and low promo dependency, indicating genuine brand pull rather than discount-driven demand
• Category funnel: showing which product categories are associated with low purchase history versus high purchase history, as a proxy for entry-point versus retention categories

4) Retention Playbook (Business Recommendations)

Translate your findings into two outputs the brand can act on. Every recommendation must state the trade-off clearly — not just what to do, but what you risk by doing it.
Promotional sunset plan: Identify which segments to gradually stop discounting, why those segments specifically, and what the margin impact looks like. The recommendation must name the segment, the trigger behavior, the rollout timeline, and the metric you would track to know if it is working. "Reduce discounts" is not an acceptable answer.
Ideal customer profile: A data-backed description of the brand's most valuable customer type, specific enough that a marketing team could use it to make targeting decisions today.

Dataset

Link

The Central Analytical Challenge

The Core Constraint

The dataset has no loyalty score, no churn label, and no timestamps. Every concept you use must be constructed from available variables, not assumed.

Loyalty must be defined, not declared. You are required to build at least two competing definitions using different variable combinations, test both, and argue clearly for one. Your justification must be grounded in internal consistency, correlation with revenue, or predictive logic — not intuition.

Every segment claim must be traceable. Labels like "at-risk" or "high-value" are only valid if they map to a specific, stated combination of variables. Assertions that cannot be traced back to the data will not meet the standard of this project.

Expected Outcome

A comprehensive customer intelligence report and interactive dashboard designed to provide the brand with a clear and actionable answer to a critical strategic question:

"Is the business successfully building a loyal customer base, or is it reliant on continuous promotional activity? Furthermore, what strategic actions should be taken under either scenario?"

Deliverables

Python	Cleaned dataset + engineered features (dependency score, value tier, satisfaction flag)
SQL	Segmentation queries answering all 5 key questions above
Power BI	Four-panel founder dashboard as described in scope
Playbook	Promo sunset plan + ideal customer profile
Summary	A concise executive summary (max 1 page) of findings and recommendations

Point of Contact

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Resources

Link

Unlocking Behavioral Intelligence in Airline Loyalty Programs

Background

Airlines use loyalty programs to increase repeat business and improve Customer Lifetime Value (CLV). However, many programs still focus mainly on points and rewards. As a result, many members stay inactive, high-value customers may stop flying without being noticed, and marketing teams often react too late. Airlines now need a more proactive and data-driven loyalty strategy.

The Challenge

You are part of the Business Analytics team of a mid-sized airline. You will work with data for around 16,700 Canadian loyalty members from 2012 to 2018.
Your goal is to build a solution that helps the marketing team identify customers likely to disengage, understand which members are most valuable, and recommend actions to improve retention. The main challenge is not only building a model, but also turning insights into clear business actions.

Core Objectives

1) Churn Prediction

The first objective is to identify members who are likely to stop engaging with the airline in the coming months.The dataset contains formal cancellation records, but cancellation is not the only form of churn that matters. Some customers may stop flying for a long period, while others may remain enrolled in the loyalty program but stop earning or redeeming points.
You are expected to define what churn means in your analysis and explain the reasoning behind your definition. There is no single correct answer, but your approach should be logical, realistic, and easy to defend.
An important challenge in this task is avoiding the use of future information. Your model should only use data that would have been available at the time of prediction. How you define churn, create features, and avoid data leakage will be an important part of the evaluation.

2) Customer Segmentation & Value

The business already tracks CLV. Your job is to interrogate whether that metric is telling the full story. What makes a customer genuinely valuable to an airline, not just historically, but going forward? How do behavioral signals (flight frequency, redemption patterns, seasonal activity) interact with demographic ones (salary band, education, marital status, card tier)? Surface the patterns that a marketing manager looking at a summary report would miss.
Your segmentation framework will be evaluated on whether it produces segments that are meaningfully distinct and actionable, and not just statistically separable.

3) Smart Retention

For each segment you identify, propose specific interventions. Vague recommendations ("offer bonus points to at-risk members") are not appreciated. A strong recommendation is one that a non-technical manager could hand to an operations team tomorrow; it would specify who receives it, when, why, and in what form.
How you structure the logic connecting churn risk to retention action is an open design problem. Elegant solutions will be recognized.

The Dataset - Link

Four integrated files:

- Customer Loyalty History — Demographics, loyalty tier, CLV, and cancellation records
- Customer Flight Activity — Monthly time-series of flights booked, distance traveled, and points accumulated/redeemed
- Calendar — Date dimension mapping months to quarters and seasons
- Data Dictionary — Full column definitions

The data is real-world in character: it contains anomalies, inconsistencies, and edge cases. Part of your analytical work is identifying these, documenting the decisions you make to address them, and ensuring those decisions do not compromise your model.

Deliverables

1) Working Prototype

A functional interface (any tool: Streamlit, Power BI, Tableau, Excel, or other) that a non-technical marketing manager can open and use without guidance. It must close the gap between your model's output and a concrete business action. What that interface looks like, and what it prioritizes, is your design call.
Standard: A first-time user should be able to identify who needs attention and what to do about it without reading a manual.

2) Technical Report (6-8 pages)

Cover: how you framed the problem, what cleaning decisions you made and why, the logic behind your model and segmentation choices, the most interesting things you found, and three business recommendations that a CFO and a CMO could both get behind. The report is written for a smart non-technical reader, not a peer reviewer.

Point of Contact

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Optimizing Delivery ETAs with Graph-Based Network Intelligence

Background

Delhivery is India's largest fully-integrated logistics provider, operating a vast network of facilities, inter-city routes, and last-mile delivery across every major state. At the core of its operations is a hub-and-spoke model: shipments travel from a source facility through one or more intermediate hubs before reaching the destination, each hop is a segment of a multi-leg journey.
To estimate delivery times, Delhivery uses OSRM, a standard routing engine that assumes clean traffic and shortest paths. But real-world logistics is far messier: congestion, facility dwell time, seasonal volume spikes, and route-type constraints all cause actual delivery times to deviate significantly from predictions.

The Challenge

The strategic question: Delhivery's OSRM system underestimates actual delivery time on a significant fraction of routes. Can a graph-based model, one that treats the logistics network as a connected graph of facilities and corridors, not a collection of independent point-to-point estimates, produce more accurate ETAs and identify which corridors and hubs are systematically causing delays?
When ETA is wrong, SLAs are missed and customers are unhappy
Downstream capacity planning breaks down across the network
There is currently no systematic way to identify which hubs and corridors are the biggest contributors to delays
Route-type decisions (FTL vs Carting) are made without accounting for graph position or structural risk of a facility

Your Mission

As a data science team, your goal is to build a graph-based intelligence system for Delhivery's logistics network. You will model the entire network as a directed graph, facilities as nodes, corridors as edges, and use this structure to produce smarter ETA predictions, surface bottleneck hubs, and generate actionable recommendations for the Head of Network Operations.
Your final output is not just a model, it is a consulting deliverable: a strategy memo that a real operations leader could act on.

What we expect from you

- Technical rigor, Clean, reproducible code with clear documentation. Model choices should be justified, not just applied
- Analytical thinking, Go beyond running the model. Interpret what the graph metrics mean for real operations. Why is a particular hub a bottleneck? What would fixing it cost vs. save?
- Business communication, The strategy memo must be written for an operations leader, not a data scientist. No raw model outputs, translate findings into decisions
- Creativity, Suggest additional insights, visualizations, or analyses beyond the provided scope if you spot something interesting in the data

Key Tasks and Deliverables

1. Graph construction & data pipeline -Parse and merge raw trip segments into a directed weighted graph where edge weights capture the median actual-vs-OSRM delay ratio per corridor, stratified by route type and time of day. Code must be clean, reproducible, and model choices justified - not just applied.
2. Bottleneck & corridor audit - Compute betweenness centrality, in/out-degree, and clustering coefficients to identify critical chokepoint hubs and chronically delayed corridors (actual time exceeds OSRM by >20%). Visualize and rank by SLA breach contribution.
3. Graph-enhanced ETA prediction model - Build and benchmark a baseline regression (trip-level features) against a GraphSAGE/node2vec-enhanced model. The graph model must demonstrably outperform the baseline on MAE and on % of trips with predicted ETA within 15% of actual. The "graph advantage" must be measured, not claimed.
4. FTL vs Carting decision framework - Build an ML-backed framework for route-type selection with the time-cost trade-off quantified for different corridor profiles, accounting for distance, time of day, and the source facility's graph position.
5. Network Operations Strategy Memo - Written for an operations leader, not a data scientist. Name the top 5 bottleneck hubs with their estimated SLA breach contribution, recommend corridor-specific interventions (parallel route, facility upgrade, or route-type shift), and quantify the % reduction in late deliveries and revenue-at-risk recovered if the top 3 hubs are upgraded. No raw model outputs - translate findings into decisions.

Expected Impact

- By the end of this project, your solution should help Delhivery's operations team:
- Make smarter ETA predictions that reduce the gap between promised and actual delivery time
- Identify and prioritize hub upgrades based on their structural risk and SLA breach contribution
- Reduce SLA breaches on chronic delay corridors through targeted interventions
- Improve route-type decisions with a data-backed FTL vs Carting framework
- Recover revenue at risk by quantifying and acting on the highest-impact bottlenecks

What Your Final Submission Should Include

1. Well-documented code, data pipeline, graph construction, model training, and benchmarking
2. Graph visualizations of the logistics network with bottleneck hubs and delay corridors highlighted
3. Model performance comparison (baseline vs. graph-enhanced) on both MAE and the 15%-accuracy business metric
4. FTL vs Carting decision framework with supporting analysis
5. Network Operations Strategy Memo (1-2 pages) with the top 5 bottleneck hubs, corridor-specific interventions, and revenue impact estimates
6. Optional: A live Streamlit dashboard showing the network with real-time delay risk scores

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Dataset & Resources

Link (Dataset) **Link (Resources)**